AI Agent

27-09-2026

Teaching Records

What we will cover

- What is AI Agent
- Prompt techniques
- Use cases of AI Agent
- Use of development tools on AI Agent
  - n8n (Low code solution)
  - Dify (Low code solution)
  - Python
- Workflow development with n8n
- AI Chatbot with Telegram and n8n
- Agentic workflow with n8n
  - Use case: Agentic Personal assistant (Google Services)
  - Overview of Personalized agent (Openclaw/Hermes) with n8n
- Intelligent chatbot with Dify and n8n
  - RAG
  - Text-to-SQL
- Other advanced topics
  - MCP
  - Multi-Agent Deployment
  - Human-in-the-loop

Class one

- Overview of AI Solutions
  - ML
    - Use case specific
      - Dynamic Customer segmentation
    - **Training data**
    - Machine learning algorithm
    - **Machine learning model**
  - LLM
    - Trained ML model
    - Generic
    - Content creation
    - Content summarization
      - Content comparison
        - E.g. CV screening
- LLM solutions
  - API
    - Charges (token)
  - Local
    - GPU

Class one

- AI use cases
  - Workflow
  - Chatbot
  - Agent
- Prompt techniques
  - Zero shot
  - Few shot
    - With Examples

Class one

- n8n
  - Deploy baseline n8n flow
  - Setup LLM connection (Azure openai)
- Installation of n8n (To be continued)
- AI workflow with n8n (To be continued)

Class two

- Types of AI Applications
  - Workflow
  - Chatbot
  - Agent
    - Follow-up
    - Validate
- Overview of Hermes agent
- Develop workflow with n8n
- Setup of telegram account in mobile

Class three

- Architect role of AI Agentic solutions
  - Solution framework
    - System integration (SI)
      - Interconnect with other systems
        - E.g. Telegram bot/Database/Google services/Microsoft services)
- Setup telegram bot
  - Token (i.e. password)
  - Chat ID
- What is Webhooking
- Webhooking with ngrok

Class exercise:

- Revise summarization engine (Add telegram node)
- Setup openrouter LLM (e.g. Deepseek v4 pro)
- AI Chatbot with Telegram

Class four

- Task automation with n8n and **Google Services**
- Create google API
- Create workflow with google service (Gmail)
- **Agentic workflow** with n8n
- **Tools calling** and **MCP**
- Overview of AI coding (Vibe coding)
- Web scraping with Python (and AI Agents) (To be continued)
- Document searching (RAG) (To be continued)